Apple //e — Custom ICs

MMU 344-0010 · IOU 344-0020 · AY-5-3600 KEYBOARD

DIP pinouts from the top, notch up, pin 1 top-left. Power **red**, ground grey, clocks blue. Verify against the datasheet before powering up.

Legend — reading the pinouts
notation

`/NAME`	**Active LOW** — true / asserted when the pin is at logic **LOW** (~0 V). A leading **/** (datasheets often use an overline) marks these: e.g. `/RES` `/IRQ` `/CS` `/WE`.
`NAME`	**Active HIGH / normal** — asserted when the pin is **HIGH** (~+5 V). No slash. `R/W` is a level: HIGH = read, LOW = write.

Colour key: **Vcc / Vdd = power** · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

MMU 344-0010 mem mgmt //e
40-DIP

GND	1	40	A1
A0	2	39	A2
PHI0	3	38	A3
Q3	4	37	A4
/PRAS	5	36	A5
ORA0	6	35	A6
ORA1	7	34	A7
ORA2	8	33	A8
ORA3	9	32	A9
ORA4	10	31	A10
ORA5	11	30	A11
ORA6	12	29	A12
ORA7	13	28	A13
R/W	14	27	A14
/INH	15	26	A15
/DMA	16	25	Vcc
/EN80	17	24	CXXXOUT
/KBD	18	23	/CASEN
/ROMEN2	19	22	R/W245
/ROMEN1	20	21	MD7

//e Memory Management Unit (344-0010): bank/aux-memory switching, language card, /INH, /DMA, address decode. Pinout from the frozen-signal reverse-engineering project. **Fails:** dead //e, no aux RAM / 80-col, bad language card.

IOU 344-0020 I/O unit //e
40-DIP

GND	1	40	H0
/LGRTXT	2	39	/SYNC
SEGA	3	38	/WNDW
SEGB	4	37	/CLRGAT
SEGC	5	36	/RA10
/80COL	6	35	/RA9
CASSO	7	34	VID6
SPKR	8	33	VID7
MD7	9	32	IKSTRB
AN0	10	31	IAKD
AN1	11	30	/C0XX
AN2	12	29	A6
AN3	13	28	Vcc
R/W	14	27	Q3
/RESET	15	26	PHI0
NC	16	25	/PRAS
ORA0	17	24	ORA7
ORA1	18	23	ORA6
ORA2	19	22	ORA5
ORA3	20	21	ORA4

//e I/O Unit (344-0020): video timing (/SYNC, /WNDW, segments), soft switches, speaker, annunciators, keyboard strobe, cassette. Holds /RESET (open-drain). **Fails:** no/bad video, dead keys, no speaker.

AY-5-3600 keyboard enc.
40-DIP

Option	1	40	X0
Option	2	39	X1
Option	3	38	X2
Option	4	37	X3
Option	5	36	X4
B9	6	35	X5
B8	7	34	X6
B7	8	33	X7
B6	9	32	X8
B5	10	31	DlyNode
B4	11	30	VCC
B3	12	29	ShiftIn
B2	13	28	CtrlIn
B1	14	27	VGG
VDD	15	26	Y9
DataRdy	16	25	Y8
Y0	17	24	Y7
Y1	18	23	Y6
Y2	19	22	Y5
Y3	20	21	Y4

Keyboard encoder on the //e and //c. Scans the X/Y matrix into a 10-bit code (B1-B9 + DataRdy). PMOS: VDD(15)/VCC(30)/VGG(27); pins 1-5 are option straps. **Fails:** dead/stuck keyboard.

//e notes
info

`MMU 344-0010`	main board, near the CPU
`IOU 344-0020`	next to the MMU
`ORA0-7`	shared MMU↔IOU multiplexed addr
`MD7`	shared bank/soft-switch data bit
`Enhanced //e`	65C02 + new ROMs; same MMU/IOU

MMU + IOU together replace the discrete glue of the][/][+. They share the ORA0-7 and MD7 lines.

Test & Repair
tips

`Dead / no boot`	MMU / ROM / 65(C)02
`No 80-col / aux`	MMU (+ aux RAM)
`No / bad video`	IOU / char ROM
`Dead keys`	AY-5-3600 / IOU strobe
`No speaker`	IOU

MMU faults kill banking (dead, or no aux memory); IOU faults hit video, keyboard strobe, and sound.

Notes
⚠ verify these

`MMU / IOU source`	frozen-signal reverse-eng project
`IOU pin 28`	Vcc (doc typo listed it as 27)

Amber pins and ⚠ **flagged chips** are my best reconstruction from incomplete/garbled sources — confirm each against a datasheet before use.